

$$= \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \text{??}$$

8. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first reflects points through the horizontal x_1 -axis and then reflects points through the line $x_2 = x_1$.

Solution: Transformation 1: $T(e_1) = e_1$ $T(e_2) = -e_2$

Transformation 2: $T(e_1) = e_2$ $T(-e_2) = -e_1$

$\therefore T(e_1) = e_2$ $T(e_2) = -e_1$

By Theorem 10, The standard matrix is $[T(e_1) \ T(e_2)] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

9. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first performs a horizontal shear that transforms e_2 into $e_2 - 2e_1$ (leaving e_1 unchanged) and then reflects points through the line $x_2 = -x_1$.

Solution: Transformation 1: $T(e_2) = e_2 - 2e_1$ $T(e_1) = e_1$

Transformation 2: $T(e_1) = -e_2$ $T(e_2 - 2e_1) = 2e_2 - e_1$

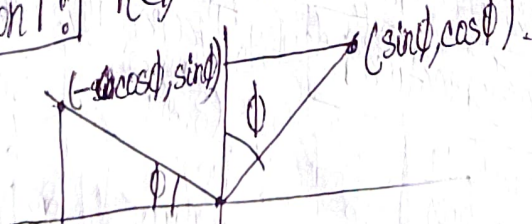
By Theorem 10, The standard matrix is: $[T(e_1) \ T(T(e_2))]$
 $= \begin{bmatrix} -e_2 & 2e_2 - e_1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 2 \end{bmatrix}$

10. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first reflects points through the vertical x_2 -axis and then rotates points $\pi/2$ radians.

Solution: Transformation 1: $T(e_1) = -e_1$ $T(e_2) = e_2$

Transformation 2:

$T(e_1) = \begin{bmatrix} -\cos \phi \\ \sin \phi \end{bmatrix}$
 $T(e_2) = \begin{bmatrix} \sin \phi \\ \cos \phi \end{bmatrix}$



By Theorem 10, the standard matrix is:
 $[T(T(e_1)) \ T(T(e_2))]$
 $= \begin{bmatrix} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$